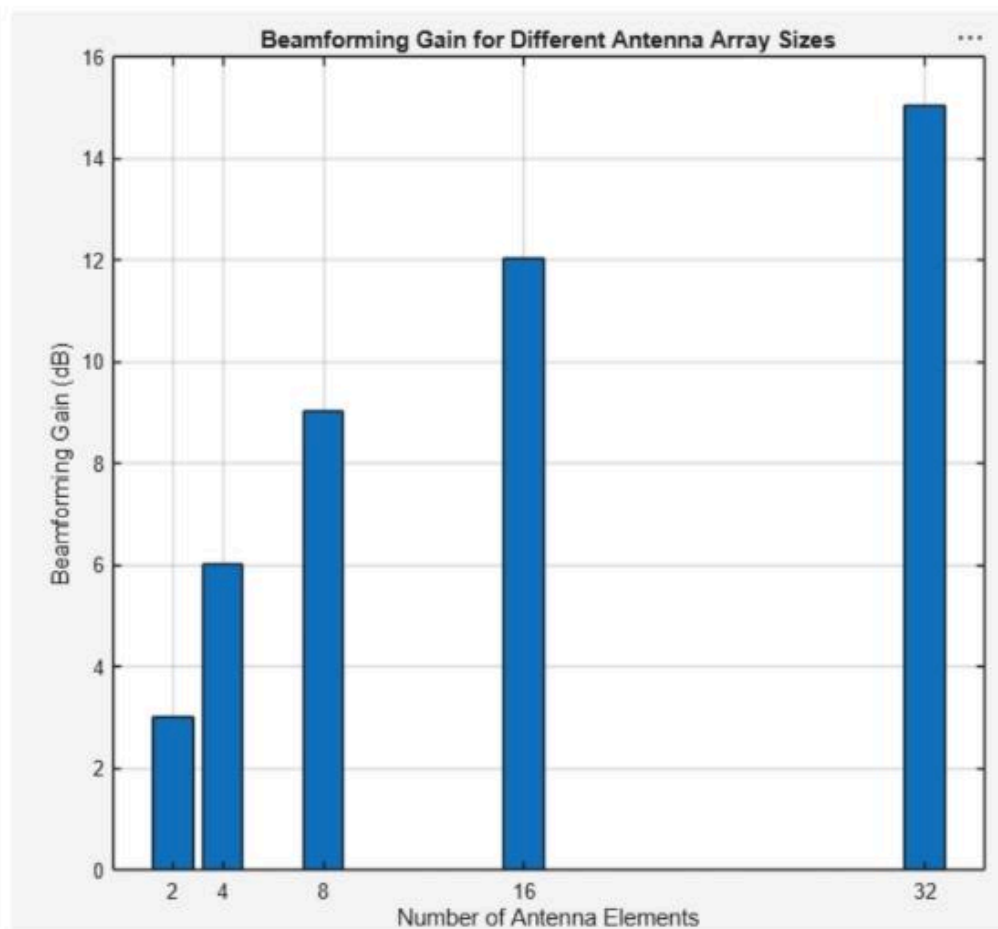


Ex.No.7: Analog Beamforming Analysis Using MATLAB

Objective 1: Matlab Code and Output

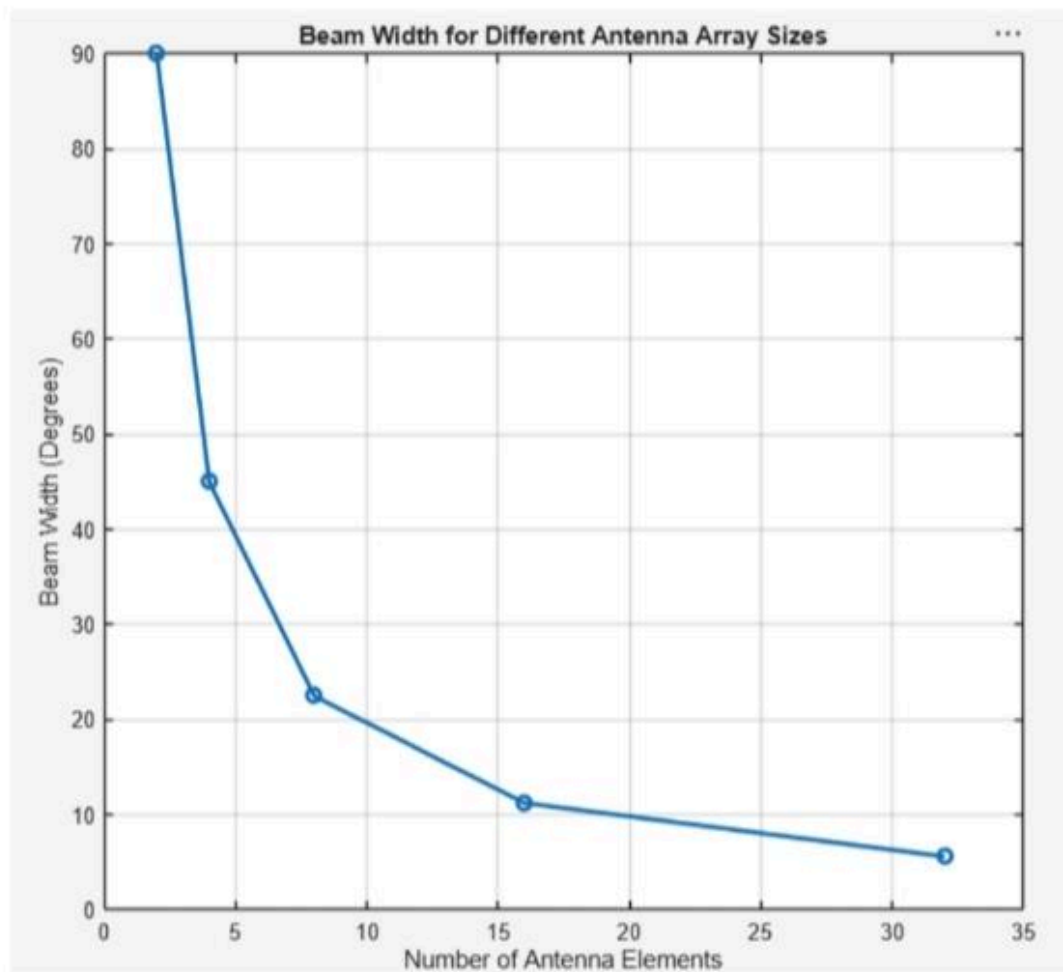
/MATLAB Drive/slotb7.m

```
1   clc;
2   clear;
3   close all;
4   antennas = [2 4 8 16 32];
5   gain = 10*log10(antennas);
6   figure
7   bar(antennas, gain)
8   grid on
9   % Number of Antenna Elements
10  % Beamforming Gain (dB)
11  xlabel('Number of Antenna Elements')
12  ylabel('Beamforming Gain (dB)')
13  title('Beamforming Gain for Different Antenna Array Sizes')
14
```



Objective 2: Matlab Code and Output

```
15 antennas = [2 4 8 16 32];  
16 % Number of Antenna Elements  
17 beamwidth = [90 45 22.5 11.25 5.625];  
18 figure  
19 % Beam Width (degrees)  
20 plot(antennas, beamwidth, 'o-', 'LineWidth', 2)  
21 grid on  
22 xlabel('Number of Antenna Elements')  
23 ylabel('Beam Width (Degrees)')  
24 title('Beam Width for Different Antenna Array Sizes')  
25
```



Objective 3: Matlab Code and Output

```
26 theta = [-60 -30 0 30 60];  
27 gain = [1 1 1 1 1];  
28 figure  
29 % Main Lobe Direction (degrees)  
30 % Constant normalized gain  
31 polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)  
32 title('Main Lobe Direction in Analog Beamforming')
```

